

Ethan Niu

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EDUCATION

University of Waterloo

Sep 2026 – Apr 2031

Systems Design Engineering

Waterloo, ON

St. Robert Catholic High School

Sep 2022 – May 2026

International Baccalaureate (IB) Diploma & Ontario Secondary School Diploma (OSSD)

Markham, ON

EXPERIENCE

Rink Guard

Jan 2026 - Apr 2026

City of Toronto

Toronto, ON

- Supervised public skating sessions, monitoring rink conditions and skater conduct to ensure a safe environment
- Stayed alert for hazards and rule violations, responding quickly to incidents and enforcing facility safety protocols during high-traffic sessions

Jul 2025 – Aug 2025

Royal Ontario Museum (ROM) Toronto, ON

- Helped coordinate children's learning through art and exhibit viewings
- Stayed alert for potential problems with children and resolved problems quickly and cleanly

PROJECTS

Toronto Accessibility Heatmap – Personal Project

July 2026 – Present

- Built a 15-minute walking-accessibility model of Toronto in QGIS, generating heatmaps for five amenity types (community centres, schools, subway stations, supermarkets, hospitals) and combining them into a weighted composite accessibility score
- Currently overlaying 2021 Census Dissemination Area population data onto the composite score to identify densely populated, under-served areas of the city, to recommend where future amenities and transit infrastructure should be prioritized

Toronto Subway System Redesign – Personal Project

Jul 2025 – Aug 2025

- Proposed a redesign of Toronto's subway network informed by bus and streetcar ridership data, planned future infrastructure expansions, and environmental impact considerations

NextStop – Partner Project

Aug 2026 – Present

- Building an interactive web app that models Toronto transit-expansion proposals, using a transparent, rule-based scenario engine (not a trained ML model) to select up to three corridor upgrades (bus priority, BRT, light rail, or metro) against user-defined budget, frequency, and priority tradeoffs
- Modeling real TTC route to score competing proposals on projected time savings, network connectivity, and cost, informed by international transit case studies (Tokyo, Seoul, Beijing)

TECHNICAL SKILLS

Software & Tools: QGIS

Programming: Fundamental Java and C++

Languages: English, French, Chinese

ACTIVITIES

DECA

- Provincial Finalist x3, competed in marketing business case competitions and roleplay events, delivering live presentations and pitches to judges under time restraints